GRAVITYSORT · USER REQUIREMENTS

GravitySort user requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	GVS-URS-001	Version	0.1
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gravitysort		
Source	docs/03a-user-requirements.md @ 6c7e5bc		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.8 pipeline): user needs drawn from the users and stakeholders in GVS-PRB-001 Tables 1 to 3 and from the decisions in GVS-DEC-001; proposed for review with the co-design partner

GravitySort user requirements

What the people who would use, build and support GravitySort need, written from their side. Product requirements that meet these needs are in GVS-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix GVS-RTM-001 (`03b-traceability.md`). Method: STANDARDS section 22. None of these needs has yet been confirmed by the users themselves: they are drawn from the problem statement, the cited literature and Amish's decisions, and are proposals for the first co-design sessions with the partner in Colombia (GVS-DEC-001, 2026-10-02).

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Regulatory	No mercury	We can recover our gold without mercury at any step, so we can sell to buyers and meet the law where mercury is banned	High	Small mining group; NGO or extension officer; regulators (PRB Tables 2 and 3)
UR-002	Performance	Daily ore	It keeps up with the ore our group mills in a working day, about 1.5 t	High	Small mining group; processing center operator (PRB Table 2)
UR-003	Performance	Fine gold	It catches the fine gold that our sluices and pans lose, so we earn more from the same ore	High	Small mining group (PRB Table 2)
UR-004	Performance	Smeltable concentrate	The final concentrate is small and clean enough (about 100 g a day) to smelt with borax ourselves	High	Small mining group; women processors (PRB Table 2)
UR-005	Usability	No grid power	It runs where there is no power, by human effort that a crew can share over a shift	High	Small mining group at remote pits (PRB Table 2)
UR-006	Usability	Motor where power exists	Where we have a battery or generator, a small motor can run it instead of the pedals	Medium	Processing center operator (PRB Table 2)
UR-007	Environment and durability	Little and dirty water	It works with the little water we have, recirculated from a settling pond and carrying silt	High	Small mining group; regulators (PRB Tables 2 and 3)
UR-008	Cost	Affordable	A small group can afford the parts and earn them back in weeks	High	Small mining group; NGO or cooperative (PRB Tables 2 and 3)
UR-009	Deployment and assembly	Local build and repair	A welder in our mining town can build and repair it with the tools they have, without a lathe	High	Local workshop or welder (PRB Table 2)
UR-010	Logistics	Carried to site	Two people can carry it to the pit in loads and put it together with hand tools	Medium	Small mining group (PRB Table 2)
UR-011	Safety	Safe to run	The spinning and moving parts are guarded, it stops quickly, it cannot run dangerously fast unnoticed, and it does not tip over	High	Every operator; NGO duty of care (PRB Tables 2 and 3)
UR-012	Usability	Quick, secure clean-up	We can take out the concentrate quickly without tools and lock it away from theft	High	Small mining group; women processors (PRB Table 2)
UR-013	Maintenance	Wear parts on site	The parts that wear can be replaced at the site and last a season	Medium	Local workshop; small mining group (PRB Table 2)

ID	Category	Title	Need	Priority	Source
UR-014	Usability	Bowl speed and settings shown	The operator can see and set how hard the machine is working without instruments	Medium	Small mining group; NGO measuring recovery (PRB Table 2)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.

Notes

- UR-002 follows the restatement of R2 by Amish on 2026-10-02 (200 kg/h of feed time, about 1.55 t per eight-hour shift; GVS-DEC-001).
- UR-010 follows the restatement of R11 by Amish on 2026-10-01 (100 kg in all, every load 30 kg or less; GVS-DDR-003, A1).
- UR-011 includes the tipping criterion decided by Amish on 2026-10-03 (a 10 degree slope in any direction with a full tank; GVS-DEC-001).
- The value-engineering target behind UR-008 is USD 455 in parts (`project.yaml`), a hypothetical control target, not a limit (Amish, 2026-10-01).